SU($n+1$) Toda Systems on Riemann Surfaces

Chunhui Wei*

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Abstract

We survey existence and classification results for the SU($n+1$) Toda system on Riemann surfaces. This survey is intended as a brief introduction for readers interested in the subject.

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*Zhejiang University.
Email: chunhuiwei@zju.edu.cn.
Homepage: https://chunhuiwei736.github.io/.

1 The system and the solution–curve correspondence

1.1 The intrinsic curvature equation and singular divisors

Let X be a Riemann surface and $\Sigma \subset X$ a closed discrete set. Fix $n \geq 1$ and define $N = n+1$. Consider an n-tuple $(g_1, \dots, g_n)$ of smooth Riemannian metrics on $X^\circ = X \setminus \Sigma$, each compatible with the complex structure of X. Denote their area forms by $\boldsymbol{\omega} = (\omega_1, \dots, \omega_n)^t$. These are positive $(1,1)$-forms and hence Kähler forms. Denote the Gaussian curvature of g_i by K_i and adopt the Ricci-form convention

$$\mathrm{Ric}(\omega_i) = K_i \omega_i, \qquad \mathrm{Ric}(\omega_i) = -\sqrt{-1}\,\partial\bar{\partial} \log h_i \quad \text{if } \omega_i = \frac{\sqrt{-1}}{2} h_i \, dz \wedge d\bar{z}.$$

The Ricci form is globally defined and does not depend on the coordinate used in the second expression.

Define the matrix

$$A = (a_{ij})_{1 \leq i,j \leq n}, \qquad a_{ii} = 2, \quad a_{i,i+1} = a_{i+1,i} = -1,$$

all other entries being zero. The $\mathrm{SU}(n+1)$ *Toda system* is the curvature equation

$$\mathrm{Ric}(\boldsymbol{\omega}) = \frac{1}{2} A \boldsymbol{\omega}, \qquad \text{equivalently} \qquad K_i \omega_i = \frac{1}{2} \sum_{j=1}^{n} a_{ij} \omega_j \quad \text{on } X^\circ. \tag{1.1}$$

Thus, with $\omega_0 = \omega_N = 0$, its ith equation reads $K_i \omega_i = \omega_i - \frac{1}{2}\omega_{i-1} - \frac{1}{2}\omega_{i+1}$. For $n = 1$, this reduces to $K_1 = 1$. In higher rank, the individual curvatures need not be constant. No background metric is needed to state this system [4, 5]. It is also called the *open Toda system* [6].

To prescribe conical singularities, fix real divisors

$$D_i = \sum_{p \in \Sigma} \gamma_{i,p} p, \qquad \gamma_{i,p} > -1, \qquad [D_i] = \sum_{p \in \Sigma} \gamma_{i,p} [p],$$

where $[p]$ is the Dirac $(1,1)$-current of total mass one. A metric represents D_i if, near each p, in a coordinate centered there,

$$\omega_i = \frac{\sqrt{-1}}{2} |z|^{2\gamma_{i,p}} \mathrm{e}^{\phi_i(z)} dz \wedge d\bar{z}, \qquad \phi_i \text{ continuous at } 0.$$

Its cone angle is $2\pi(1 + \gamma_{i,p})$, and its area is locally finite. Zero weights are allowed. The divisors are effective and integral only when all their weights are nonnegative integers. The singular system is the global equality of currents

$$\mathrm{Ric}(\omega_i) + 2\pi [D_i] = \frac{1}{2} \sum_{j=1}^{n} a_{ij} \omega_j \quad \text{on } X, \qquad i = 1, \dots, n. \tag{1.2}$$

Here the Ricci current is defined using the locally integrable logarithm of the metric coefficient. Its atom at p is $-2\pi\gamma_{i,p}[p]$. In particular, $\mathrm{Ric}(\omega_i) = K_i \omega_i$ is the smooth identity on X°. The divisor term records its distributional extension across Σ.

Normalization and coordinate expressions. The geometric convention in [4, 5] uses the normalization $\mathrm{Ric}(\widehat{\boldsymbol{\omega}}) = 2A\widehat{\boldsymbol{\omega}}$. It agrees with (1.1) after the constant rescaling $\widehat{\omega}_i = \omega_i/4$, since the Ricci form is unchanged by constant rescaling. With $[p]$ normalized to have total mass one, the singular equation in that normalization is $\mathrm{Ric}(\widehat{\omega}_i) + 2\pi[D_i] = 2\sum_j a_{ij}\widehat{\omega}_j$.

Define U_i by $\omega_i = \frac{\sqrt{-1}}{2} e^{U_i} dz \wedge d\bar{z}$, and define the Euclidean Laplacian by $\Delta = \partial_x^2 + \partial_y^2 = 4\partial_z \partial_{\bar{z}}$. The current equation (1.2) has the local form

$$\Delta U_i + \sum_{j=1}^{n} a_{ij} e^{U_j} = 4\pi \sum_{p \in \Sigma} \gamma_{i,p} \delta_p, \qquad i = 1, \dots, n, \tag{1.3}$$

where δ_p now denotes the scalar Dirac distribution relative to Euclidean area. The conical behavior is

$$U_i(z) = 2\gamma_{i,p} \log |z| + O(1). \tag{1.4}$$

Under $z = z(\zeta)$, $U_i^{\zeta} = U_i^z \circ z + \log |dz/d\zeta|^2$: the U_i are logarithms of local metric coefficients. The alternative local convention $\partial_z \partial_{\bar{z}} u_i + \sum_j a_{ij} e^{u_j} = \pi \sum_p \gamma_{i,p} \delta_p$ is related to (1.3) by $U_i = u_i + \log 4$.

For analytic calculations, one may choose a smooth conformal background metric g on X and define v_i by $g_i = e^{v_i} g$, or, equivalently, $\omega_i = e^{v_i} dV_g$. Equation (1.2) then becomes

$$\Delta_g v_i + \sum_j a_{ij} e^{v_j} = 2K_g + 4\pi \sum_{p \in \Sigma} \gamma_{i,p} \delta_p, \tag{1.5}$$

with scalar Dirac distributions relative to dV_g. The inverse Cartan matrix is

$$a^{ij} := (A^{-1})_{ij} = \frac{\min(i,j)\big(N - \max(i,j)\big)}{N}. \tag{1.6}$$

1.2 Local correspondence and the infinitesimal Plücker formula

Let $\Omega \subset \mathbb{C}$ be simply connected. A holomorphic curve $f : \Omega \to \mathbb{CP}^n$ is *linearly full*, also called *nondegenerate*, if its image is contained in no projective hyperplane. For a local holomorphic lift $F = (F_0, \dots, F_n)$, define

$$\tau_0 = 1, \qquad \tau_k = \|F \wedge F' \wedge \cdots \wedge F^{(k-1)}\|^2, \quad 1 \le k \le N. \tag{1.7}$$

The norm is induced by the standard Hermitian form on $\mathbb{C}^N$. The curve f is *totally unramified* on Ω when $W(F_0, \dots, F_n) \neq 0$ there. Equivalently, all its osculating wedge vectors in (1.7) are nonzero.

Theorem 1.1 ([1, 2, 3])**.** *Every smooth solution of* (1.3) *with* $\Sigma = \varnothing$ *on* Ω *arises from a linearly full, totally unramified holomorphic curve by*

$$e^{U_i} = 4 \frac{\tau_{i-1} \tau_{i+1}}{\tau_i^2}, \qquad 1 \le i \le n. \tag{1.8}$$

Conversely, every such curve gives a smooth solution. Two curves give the same solution exactly when they differ by a constant element of $\mathrm{PSU}(N)$*.*

Multiplying the lift F by a nowhere-zero holomorphic function leaves (1.8) unchanged. One may normalize its Wronskian to be 1 on a simply connected domain. Then $\tau_N = 1$ and

$$U_i = \log 4 - \sum_{j=1}^{n} a_{ij} \log \tau_j. \tag{1.9}$$

For SU(2), this is Liouville's formula

$$e^{U_1} = \frac{4|f'|^2}{(1 + |f|^2)^2}$$

in an affine target coordinate.

1.3 Global correspondence and monodromy

On a surface that is not simply connected, the holomorphic curve associated with a solution is generally multivalued. In [3, 4, 5], a linearly full multivalued holomorphic curve with monodromy in $\mathrm{PSU}(N)$ is called a *unitary curve*. Let $X^\circ = X \setminus \Sigma$. Such a curve is described on the universal cover by

$$f : \widetilde{X^\circ} \longrightarrow \mathbb{CP}^n, \qquad \sigma : \pi_1(X^\circ) \longrightarrow \mathrm{PSU}(N), \qquad f(\alpha \cdot z) = \sigma(\alpha) f(z). \tag{1.10}$$

The curve is totally unramified away from the prescribed sources. For $0 \le k \le n-1$, let f_k be its kth associated (osculating) curve in $\mathbb{P}(\Lambda^{k+1}\mathbb{C}^N)$, locally represented by $F \wedge F' \wedge \cdots \wedge F^{(k)}$. Normalize the Fubini–Study form on each projective space by $\omega_{\mathrm{FS}} = 2\sqrt{-1}\,\partial\bar\partial \log(1+\sum_a |w_a|^2)$ in affine coordinates. Its restriction to a projective line has Gaussian curvature one. The intrinsic form of the solution–curve correspondence is

$$(\omega_1, \dots, \omega_n) = (f_0^*\omega_{\mathrm{FS}}, \dots, f_{n-1}^*\omega_{\mathrm{FS}}). \tag{1.11}$$

These forms are first defined on $\widetilde{X^\circ}$. Projective unitary monodromy preserves ω_{FS}, so these forms descend to X°, where they satisfy (1.1). Formula (1.8) is their local coordinate expression [1, 2, 4].

Theorem 1.2 ([3, 4, 5])**.** *Tuples of metrics satisfying the curvature system* (1.1) *on* X° *correspond, through* (1.11)*, to equivariant, totally unramified holomorphic curves* (1.10)*, modulo the action* $(f, \sigma) \mapsto (g \circ f, g\sigma g^{-1})$ *of* $g \in \mathrm{PSU}(N)$*. Conical extension across a marked point corresponds to a* regular singularity *of the unitary curve, described by the local normal form of Theorem 2.1.*

The associated curve is single-valued precisely when its projective monodromy is trivial. The correspondence uses a $\mathrm{PSU}(N)$ representation and requires no lift to $\mathrm{SU}(N)$.

At a source, the cone weights determine the successive differences of the local exponents, and hence the eigenvalue ratios of the local monodromy. Nonnegative integer weights correspond to ordinary osculating ramification and trivial *local projective* monodromy. Even if the local projective monodromy is trivial at every source, monodromy around the handles of X can remain nontrivial.

2 Explicit formulas around a singular source

2.1 Exponents and the holomorphic normal form

Consider a solution on a disk whose only source is at 0, with weights $\gamma_i > -1$ and finite area in every component near 0. Define

$$\eta_i = \sum_{j=1}^{n} a^{ij}\gamma_j, \qquad \beta_0 = -\eta_1, \quad \beta_i = \eta_i - \eta_{i+1} + i \ (1 \le i < n), \quad \beta_n = \eta_n + n. \tag{2.1}$$

These numbers satisfy

$$\beta_i - \beta_{i-1} = 1 + \gamma_i > 0, \qquad \sum_{a=0}^{n} \beta_a = \frac{nN}{2}. \tag{2.2}$$

Theorem 2.1 ([3, Theorem 1.2])**.** *After a unitary change of target coordinates, a normalized holomorphic lift of the associated curve has the form*

$$F(z) = (z^{\beta_0} g_0(z), \dots, z^{\beta_n} g_n(z)), \qquad W(F_0, \dots, F_n) \equiv 1, \tag{2.3}$$

where the g_a are holomorphic and $g_a(0) \neq 0$. Conversely, holomorphic data satisfying these conditions yield a Toda solution through (1.8), with the prescribed weights and finite area on each smaller disk. The remainders $U_i - 2\gamma_i \log|z|$ extend Hölder continuously across 0.

The powers in (2.3) are taken on the universal cover of the punctured disk. Analytic continuation once around 0 multiplies F by $\operatorname{diag}(\mathrm{e}^{2\pi\sqrt{-1}\beta_0}, \dots, \mathrm{e}^{2\pi\sqrt{-1}\beta_n})$, which is unitary. In particular, the metric coefficients are single-valued. This diagonal form has no logarithmic terms, even when exponent differences are integers.

2.2 Wronskian formulas for the metric coefficients

For a subset $I \subset \{0, \dots, n\}$ of cardinality k, define

$$s_I = \sum_{a \in I} \beta_a, \qquad s_k^0 = \sum_{a=0}^{k-1} \beta_a, \qquad G_I(z) = z^{\,k(k-1)/2 - s_I} W(F_a : a \in I),$$

with indices in increasing order. Each G_I is holomorphic. Define

$$R_k(z) = \sum_{|I|=k} |z|^{2(s_I - s_k^0)} |G_I(z)|^2 \quad (1 \le k \le n), \qquad R_0 = R_N = 1. \tag{2.4}$$

Since the exponents are strictly ordered, all powers in this sum are nonnegative, and the term indexed by $I = \{0, \dots, k-1\}$ has a nonzero value at 0. In the notation of (1.7),

$$\tau_k = |z|^{-2\eta_k} R_k, \qquad 1 \le k \le n.$$

The explicit source formula is therefore

$$\mathrm{e}^{U_i(z)} = 4|z|^{2\gamma_i} \frac{R_{i-1}(z) R_{i+1}(z)}{R_i(z)^2}. \tag{2.5}$$

The factor $|z|^{2\gamma_i}$ specifies the cone angle. The positive quotient determines the bounded part of U_i through the holomorphic functions and their Wronskians [3].

For example, in the SU(3) case, define

$$T_1 = |F_0|^2 + |F_1|^2 + |F_2|^2, \qquad T_2 = \sum_{0 \le a < b \le 2} |F_a F_b' - F_b F_a'|^2.$$

For $W(F_0, F_1, F_2) = 1$, the complete local formula is

$$\mathrm{e}^{U_1} = \frac{4T_2}{T_1^2}, \qquad \mathrm{e}^{U_2} = \frac{4T_1}{T_2^2}. \tag{2.6}$$

Here $(\beta_0, \beta_1, \beta_2) = (-(2\gamma_1 + \gamma_2)/3,\ 1 + (\gamma_1 - \gamma_2)/3,\ 2 + (\gamma_1 + 2\gamma_2)/3)$.

2.3 Holomorphic coordinate normalization

After a holomorphic change of source coordinate fixing 0, Theorem 2.1 expresses the germ of the metric tuple using the fixed exponents and $n-1$ holomorphic functions nonvanishing at 0.

For $n = 1$, no free holomorphic functions remain after this coordinate normalization. With $\mu = 1 + \gamma > 0$, the metric takes the form

$$\mathrm{e}^U |dz|^2 = \frac{4\mu^2 |\xi|^{2\mu-2}}{(1 + |\xi|^{2\mu})^2} |d\xi|^2. \tag{2.7}$$

Local solvability holds for every weight vector whose entries are greater than -1.

3 Solutions on $\mathbb{C}$ with control at infinity

3.1 Finite energy: smooth solutions and a single singular source

The basic global condition is

$$M_i := \int_{\mathbb{C}} \mathrm{e}^{U_i}\,\mathrm{d}x\,\mathrm{d}y < \infty, \qquad 1 \le i \le n. \tag{3.1}$$

This *finite-energy condition* [6, 7] requires every component metric to have finite total area M_i.

Theorem 3.1 ([6])**.** *Every smooth solution on $\mathbb{C}$ satisfying (3.1) comes from a projective transform of the rational normal curve $[1 : z : \cdots : z^n]$. All such transforms yield finite-energy solutions. For a fixed coordinate on $\mathbb{C}$, their parameter space is the symmetric space $\mathrm{PSL}(N,\mathbb{C})/\mathrm{PSU}(N)$, of real dimension $n(n+2)$, and*

$$M_i = 4\pi i(N-i), \qquad U_i(z) = -4\log|z| + O(1) \quad (|z| \to \infty). \tag{3.2}$$

Concretely, the parameter space is $\mathrm{PSU}(N)\backslash\mathrm{PSL}(N,\mathbb{C})$. Inversion identifies it with $\mathrm{PSL}(N,\mathbb{C})/\mathrm{PSU}(N)$. A symmetric representative is

$$\mathrm{e}^{U_i(z)} = \frac{4i(N-i)}{(1+|z|^2)^2}.$$

Theorem 3.2 ([7])**.** *For every $\gamma_i > -1$ there are finite-energy solutions of*

$$\Delta U_i + \sum_j a_{ij}\mathrm{e}^{U_j} = 4\pi\gamma_i\delta_0 \quad \text{on } \mathbb{C}.$$

All solutions are given by an explicit finite-dimensional family. Their masses and asymptotic behavior satisfy

$$\sum_j a_{ij}M_j = 4\pi(2+\gamma_i+\gamma_{N-i}), \tag{3.3}$$

$$U_i(z) = -(4+2\gamma_{N-i})\log|z| + O(1) \quad (|z| \to \infty). \tag{3.4}$$

If no consecutive sum $\sum_{k=j}^{\ell}(1+\gamma_k)$, $1 \le j \le \ell \le n$, is an integer, every solution is radial about 0.

An explicit version of the classification uses the exponents (2.1). Choose positive $\lambda_0, \dots, \lambda_n$ and complex coefficients c_{ab}, $0 \le b < a \le n$, and define

$$F_a(z) = \sqrt{\lambda_a}\left(z^{\beta_a} + \sum_{b<a} c_{ab}z^{\beta_b}\right), \qquad 0 \le a \le n. \tag{3.5}$$

The sum is empty for $a = 0$. With the normalization $W(F) = 1$, the constraints are

$$c_{ab} = 0 \text{ if } \beta_a - \beta_b \notin \mathbb{Z}, \qquad \prod_{a=0}^{n}\lambda_a = \left(\prod_{0\le a<b\le n}(\beta_b - \beta_a)\right)^{-2}. \tag{3.6}$$

Inserting (3.5) into (1.8) gives every finite-energy one-source solution. These formulas are the Lin–Wei–Ye parametrization rewritten as a lift with $W(F) = 1$. The numerical constraint therefore differs from the one for the weighted variable $\mathrm{e}^{-\sum_j a^{1j}U_j}$ used in their paper.

The choice $c_{ab} = 0$ for all indices gives solutions for arbitrary admissible weights in this explicit family. The resonance condition $\beta_a - \beta_b \in \mathbb{Z}$ permits additional complex parameters and nonradial solutions. The real dimension is

$$n + 2\,\#\{(a,b) : 0 \le b < a \le n,\ \beta_a - \beta_b \in \mathbb{Z}\}. \tag{3.7}$$

It is n in the nonresonant case and $n(n+2)$ when every γ_i is a nonnegative integer.

Lin–Wei–Ye also prove nondegeneracy: the bounded kernel of the linearized system consists precisely of the infinitesimal variations of this family. These classifications describe the finite-energy limits of *fully bubbling* sequences, in which all components survive a common rescaling [21].

3.2 Several sources and a prescribed conical singularity at infinity

For finite marked points $p_1, \dots, p_m \in \mathbb{C}$, one can instead prescribe the asymptotic condition

$$U_i(z) = -2(2 + \gamma_{i,\infty}) \log |z| + O(1), \qquad \gamma_{i,\infty} > -1. \tag{3.8}$$

In the coordinate $\zeta = 1/z$, the transformed coefficient is $U_i(1/\zeta) - 4 \log |\zeta|$. Condition (3.8) describes a conical singularity of weight $\gamma_{i,\infty}$ at ∞. Hence this plane problem is equivalent to the compact geometric problem on $\mathbb{CP}^1$ with divisors

$$D_i = \sum_{a=1}^{m} \gamma_{i,p_a} p_a + \gamma_{i,\infty} \infty. \tag{3.9}$$

The necessary mass identity is

$$\sum_j a_{ij} M_j = 4\pi \left(2 + \sum_{a=1}^{m} \gamma_{i,p_a} + \gamma_{i,\infty} \right). \tag{3.10}$$

For one finite source, Theorem 3.2 gives the necessary and sufficient relation $\gamma_{i,\infty} = \gamma_{N-i,0}$. In particular, smooth extension of the metrics across infinity forces the finite source to be removable as well. For several finite sources, the corresponding Fuchsian equation must have unitarizable monodromy, as in the three-point results of Lin–Nie–Wei [10].

The component reversal $i \mapsto N - i$ is a special feature of the one-source theorem. It is not a general relation between arbitrarily chosen finite-source weights and the weights at infinity.

3.3 Polynomial energy growth and linear differential equations

A weaker growth condition at infinity is

$$\int_{|z|<R} \mathrm{e}^{U_1}\, \mathrm{d}x\, \mathrm{d}y = O(R^K) \quad \text{for some } K > 0. \tag{3.11}$$

Condition (3.11) is called *polynomial energy growth* in [9]. It bounds the area of the first component. Eremenko [8] uses Nevanlinna theory for curves of finite order to classify smooth entire Toda solutions with this property: their associated curves admit homogeneous coordinates forming a basis of solutions of

$$y^{(N)} + P_{N-1}(z) y^{(N-1)} + \cdots + P_0(z) y = 0, \tag{3.12}$$

where the P_j are polynomials. Conversely, any fundamental system of solutions of such an equation yields, through (1.8), a smooth Toda solution with polynomial energy growth. The $y^{(N-1)}$ term can be removed by a scalar gauge transformation without changing the projective

curve. For constant-coefficient equations, the homogeneous coordinates are exponential polynomials. The equation $y^{(N)} = 0$ gives the finite-energy solutions associated with rational normal curves.

On $\mathbb{C}^*$, Zhao's preprint [9] studies polynomial energy growth at both 0 and ∞, expressed by a polynomial bound in R for the area of the first component over the annulus $R^{-1} < |z| < R$. Zhao shows that a normalized holomorphic lift of the associated curve satisfies an equation of order N whose coefficients are Laurent polynomials, that is, sums of polynomials in z and z^{-1}. These equations may have irregular singularities at 0 and ∞. Laurent-polynomial coefficients alone do not imply the unitary monodromy required for a single-valued Toda metric on $\mathbb{C}^*$.

4 Compact Riemann surfaces with singular divisors

4.1 The prescription problem and necessary conditions

Fix a compact connected Riemann surface X, distinct marked points, and an ordered tuple of real divisors $D_i = \sum_p \gamma_{i,p} p$ with $\gamma_{i,p} > -1$. A *Toda solution* is an ordered tuple of conical Kähler metrics $(\omega_1, \dots, \omega_n)$ representing $(D_1, \dots, D_n)$ and satisfying the curvature equation

$$\mathrm{Ric}(\omega_i) + 2\pi[D_i] = \frac{1}{2}\sum_j a_{ij}\omega_j \quad \text{as currents on } X.$$

Solutions are counted as metric tuples without quotienting by automorphisms of X.

Define

$$d_i = \deg D_i, \qquad b_i = \int_X \omega_i, \qquad \mathbf{d} = (d_1, \dots, d_n)^t.$$

Since $\int_X \mathrm{Ric}(\omega_i) = 2\pi\chi(X)$ when the atomic part of the Ricci current is included, the coupled Gauss–Bonnet identity is

$$A\mathbf{b} = 4\pi\big(\chi(X)\mathbf{1} + \mathbf{d}\big), \qquad b_i = 4\pi\sum_j a^{ij}\big(\chi(X) + d_j\big) > 0. \tag{4.1}$$

Thus the component areas are determined by the divisor degrees. Existence can nevertheless depend on the individual weights, the marked points, and the complex structure.

For a smooth metric tuple, all $D_i = 0$, so (4.1) forces X to be a sphere. For singular divisors, positivity of the areas is necessary but not sufficient: the two-point sphere classification already imposes further relations between the weights. Equivalently, the prescription problem asks for a unitary curve whose osculating metrics represent all the prescribed divisors.

4.2 The sphere: complete cases and three-point families

On $\mathbb{CP}^1$, every smooth Toda metric tuple is obtained from a projective transform of a rational normal curve by the metric pullback formula (1.11) (Theorem 3.1).

For divisors supported at two marked points, choose a Möbius coordinate placing them at 0 and ∞, and define the weights by $D_i = \gamma_{i,0}[0] + \gamma_{i,\infty}[\infty]$. Such conical metric tuples exist if and only if

$$\gamma_{i,\infty} = \gamma_{N-i,0} \quad (1 \le i \le n). \tag{4.2}$$

All of them are obtained by applying (1.11) to the associated curves in (3.5)–(3.6).

For divisors supported at three points, choose a coordinate placing these points at $0, 1, \infty$. Lin–Nie–Wei [10] study the metric prescription problem through a Fuchsian differential equation of order N for the associated curve. Certain weight patterns at one marked point allow a reduction to a generalized hypergeometric equation. For SU(3), these patterns include

$$(\gamma_{1,1}, \gamma_{2,1}) = (a, 0), \quad (0, a), \quad \text{or} \quad \gamma_{1,1} + \gamma_{2,1} = -1, \qquad \gamma_{i,1} > -1.$$

The remaining weights determine two lists of real hypergeometric parameters. If their fractional parts strictly interlace on $\mathbb{R}/\mathbb{Z}$, the invariant Hermitian form is positive definite. The resulting unitary curve then induces a conical Toda metric tuple with the prescribed divisors [10, Theorem 1.23]. The rank-two converse to the hypergeometric reduction requires an additional reality condition on an accessory coefficient [10, Theorem 1.25].

Local exponents determine conjugacy classes of local monodromy. Accessory parameters determine whether the whole representation preserves a positive Hermitian form.

For SU(2), the unknown is a single spherical cone metric. Troyanov [11] treats curvature prescription in a fixed conformal class. Mondello–Panov [12] give angle constraints and existence results on the sphere under their strict admissibility hypotheses, allowing the marked conformal configuration to vary.

4.3 Flat tori: existence, finite counts, and symmetry

Fix the complex torus $E_\tau = \mathbb{C}/(\mathbb{Z}+\tau\mathbb{Z})$, $\operatorname{Im}\tau > 0$, and distinct marked points $p_0, \dots, p_m$. For SU(3), prescribe effective integral divisors $D_i = \sum_{k=0}^{m} n_{i,k}p_k$, where $n_{i,k} \geq 0$ and the weights are not all zero. The unknown metric pair satisfies

$$\operatorname{Ric}(\omega_1) + 2\pi[D_1] = \omega_1 - \tfrac{1}{2}\omega_2, \qquad \operatorname{Ric}(\omega_2) + 2\pi[D_2] = \omega_2 - \tfrac{1}{2}\omega_1.$$

Define $N_i = \deg D_i$. Since $\chi(E_\tau) = 0$, the component areas are

$$b_1 = \frac{4\pi}{3}(2N_1 + N_2), \qquad b_2 = \frac{4\pi}{3}(N_1 + 2N_2). \tag{4.3}$$

The noncritical condition $N_1 \not\equiv N_2 \pmod 3$ is equivalent to both areas avoiding $4\pi\mathbb{N}$, where $\mathbb{N} = \{1, 2, \dots\}$.

Theorem 4.1 ([13])**.** *Under this noncritical condition, the number of Toda metric pairs representing (D_1, D_2) on the fixed marked torus is finite, at least one and at most*

$$\frac{1}{3 \cdot 2^{m+1}} \prod_{k=0}^{m} (n_{1,k}+1)(n_{2,k}+1)(n_{1,k}+n_{2,k}+2). \tag{4.4}$$

For divisors supported only at 0, an even solution *exists if and only if at least one of the two weights is even.*

Here *even* means $\iota^*\omega_i = \omega_i$ for both components, where $\iota(z) = -z$ [13]. For $(D_1, D_2) = (0, [0])$, the bound equals one, giving a unique metric pair on every complex torus. For $(D_1, D_2) = ([0], 3[0])$, it equals eight and both weights are odd: there are at least two metric pairs, none invariant under ι, because pullback by ι pairs distinct tuples.

Chen–Guo–Lin [14] give a higher-rank count. For SU$(n+1)$, $n \geq 2$, prescribe effective integral divisors $D_i = \sum_k n_{i,k}p_k$, not all zero, and define $N_j = \deg D_j$ and $q_i = \sum_j a^{ij}N_j = b_i/(4\pi)$. If $q_i \notin \mathbb{Z}$ for every i, conical Toda metric tuples representing these divisors exist, and their number is at most

$$\frac{\displaystyle\prod_{k=0}^{m}\prod_{j=1}^{n}\prod_{i=1}^{j}\left(\sum_{\ell=i}^{j}(n_{\ell,k}+1)\right)}{N\left(\displaystyle\prod_{j=2}^{n} j!\right)^{m+1}}. \tag{4.5}$$

In particular, the divisor tuple $(0, \dots, 0, [0])$ is represented by a unique Toda metric tuple.

The curves associated with these metric tuples satisfy ODEs whose coefficients are elliptic functions on the torus. Apparent singularities and unitary monodromy constrain the accessory parameters. When a component area lies in $4\pi\mathbb{N}$, the noncritical counting theorem no longer applies. Chen–Lin [15] obtain separate existence, symmetry, and asymptotic results for critical SU(3) families.

4.4 Existence of metrics for prescribed divisors

The rank-two results of Battaglia–Jevnikar–Malchiodi–Ruiz [19] and the higher-rank results of Lin–Yang–Zhong [20, Theorem 1.9] imply the following geometric existence theorem.

Theorem 4.2. *Let X have positive genus and let $D_i = \sum_p \gamma_{i,p} p$ be effective integral divisors. If their prescribed areas satisfy*

$$b_i = 4\pi \sum_j a^{ij} \big(\chi(X) + \deg D_j\big) > 0, \qquad b_i \notin 4\pi\mathbb{N} \quad \text{for all } i, \tag{4.6}$$

then there exists an ordered tuple of conical Kähler metrics $(\omega_1, \dots, \omega_n)$ representing $(D_1, \dots, D_n)$ and satisfying (1.2). Its component areas are b_i.

The area condition permits arbitrarily large component areas. For general real weights, compactness results involve exceptional sets depending on the weights.

Subcritical geometric criteria. On any compact Riemann surface, divisors with nonnegative real weights are represented by a Toda metric tuple if all the prescribed areas satisfy $0 < b_i < 4\pi$ [16]. For SU(3), weights in $(-1, \infty)$ are also allowed under the sharper weighted condition

$$0 < b_i < 4\pi\theta_i, \qquad \theta_i = \min\{1, \min_{p\in\Sigma}(1+\gamma_{i,p})\}, \qquad i = 1, 2,$$

where $\theta_i = 1$ for an empty marked set [17].

The auxiliary mean-field formulation. For the analytic formulation, choose a smooth conformal metric g of area one and a smooth function ψ satisfying $\Delta_g \psi = 2K_g - 4\pi\chi(X)$. The broader mean-field system is

$$\Delta_g w_i + \sum_j a_{ij}\rho_j \left(\frac{h_j \mathrm{e}^{w_j}}{\int_X h_j \mathrm{e}^{w_j}\, \mathrm{d}V_g} - 1 \right) = 4\pi \sum_p \gamma_{i,p}(\delta_p - 1), \tag{4.7}$$

where the h_i are smooth positive functions and $\rho_i > 0$. These parameters are independent in the mean-field system. The geometric curvature problem is recovered by specializing to $\rho_i = b_i$ and $h_i = \mathrm{e}^{\psi}$ for every i, with metrics

$$\omega_i = \frac{b_i \mathrm{e}^{\psi + w_i}}{\int_X \mathrm{e}^{\psi + w_i}\, \mathrm{d}V_g}\, \mathrm{d}V_g. \tag{4.8}$$

Conversely, for a Toda metric, define v_i by $\omega_i = \mathrm{e}^{v_i} \mathrm{d}V_g$. Then $w_i = v_i - \psi$ is a scalar representative. Thus mean-field solutions modulo additive constants correspond to the metric tuples.

With $-\Delta_g G_p = \delta_p - 1$ and $\int_X G_p \, \mathrm{d}V_g = 0$, define

$$u_i = w_i + 4\pi \sum_p \gamma_{i,p} G_p, \qquad \widetilde{h}_i = h_i \exp\left(-4\pi \sum_p \gamma_{i,p} G_p \right).$$

Near p, $\widetilde{h}_i$ is the product of a smooth positive factor and $|z|^{2\gamma_{i,p}}$. On the Sobolev space of functions with mean zero, the functional is

$$J_{\boldsymbol{\rho}}(\mathbf{u}) = \underbrace{\frac{1}{2} \sum_{i,j} a^{ij} \int_X \langle \nabla u_i, \nabla u_j \rangle \, \mathrm{d}V_g}_{E_A(\mathbf{u})} - \sum_i \rho_i \log \int_X \widetilde{h}_i \mathrm{e}^{u_i} \, \mathrm{d}V_g. \tag{4.9}$$

Jost–Wang [16] prove the sharp Moser–Trudinger inequality

$$4\pi \sum_i \left(\log \int_X \mathrm{e}^{u_i} \, \mathrm{d}V_g - \int_X u_i \, \mathrm{d}V_g \right) \leq E_A(\mathbf{u}) + C.$$

For smooth positive coefficients, the functional is bounded below exactly when $\rho_i \leq 4\pi$ for all i, and strict inequalities guarantee the existence of a minimizer. The same strict subcritical condition remains sufficient in the presence of nonnegative singular weights. For SU(3), the singular Moser–Trudinger inequality of Battaglia–Malchiodi [17] replaces 4π by $4\pi\theta_i$. At $\rho_i = b_i$ and $h_i = \mathrm{e}^{\psi}$, formula (4.8) turns these minimizers into Toda metrics with the prescribed divisors.

On positive-genus surfaces with nonnegative integer weights, the mean-field system admits a solution for arbitrary smooth positive coefficients h_i whenever all $\rho_i > 0$ avoid $4\pi\mathbb{N}$ [19, 20]. Applying this result at the geometric specialization gives Theorem 4.2. Compactness results for parameters outside the exceptional sets rely on the classification of local blow-up masses and estimates for the residual part of the limiting component area measures, namely the part not concentrated at blow-up points [21, 22, 20]. Partially bubbling sequences, in which some components disappear after rescaling, and bubbling at several scales require additional analysis [21].

Critical parameters and negative singular weights. At the critical values of the Moser–Trudinger inequality, a lower bound for the functional does not necessarily imply that its infimum is attained. For the regular SU(3) mean-field system (without singular sources), Jost–Lin–Wang [18] give sufficient conditions: when $\rho_1 = 4\pi$ and $0 < \rho_2 < 4\pi$, the inequality

$$\Delta_g \log h_1 + 8\pi - \rho_2 - 2K_g > 0 \quad \text{on } X$$

ensures the existence of a minimizer. For $\rho_1 = \rho_2 = 4\pi$, a sufficient condition is $\min_i \Delta_g \log h_i + 4\pi - 2K_g > 0$.

Battaglia–Malchiodi [23] establish existence and nonexistence regimes for singular SU(3) mean-field systems, including regimes in which selected weights approach -1. For prescribed singular divisors, these results apply when their hypotheses hold at $\rho_i = b_i$ and $h_i = \mathrm{e}^{\psi}$.

4.5 Global constructions and classification of special families

Reduced solutions generated by a spherical metric. Let ω be a spherical cone metric representing D, so that $\mathrm{Ric}(\omega) + 2\pi[D] = \omega$. Then the metric tuple

$$(\omega_1, \dots, \omega_n) = \big(i(N-i)\omega\big)_{i=1}^{n} \tag{4.10}$$

satisfies the Toda curvature equation with $D_1 = \cdots = D_n = D$ [24, 5]. Its associated curve is the spherical developing map followed by the rational normal map

$$[x : y] \longmapsto [x^n : \sqrt{\tbinom{n}{1}} x^{n-1} y : \cdots : y^n].$$

The symmetric-power representation $\mathrm{PSU}(2) \to \mathrm{PSU}(N)$ ensures that its osculating metrics descend to the punctured surface and extend with the stated divisors.

Shi–Wei–Xu [24, 5] classify the larger family of *reduced solutions* generated by a fixed spherical metric. Their curves are projective transforms of this composite curve for which the induced metrics remain single-valued. For each projective monodromy element, denote a unitary representative by $R(\alpha)$. A transformation represented by $T \in \mathrm{SL}(N, \mathbb{C})$ is admissible precisely when

$$R(\alpha)^* H R(\alpha) = H \quad \text{for every } \alpha, \qquad H = T^*T > 0, \quad \det H = 1. \tag{4.11}$$

The condition is independent of the scalar factors used to choose the unitary representatives. Modulo unitary transformations of the target, these invariant positive Hermitian forms parametrize the generated metric tuples, all representing $(D, \dots, D)$. The classification in [5, Theorem 1.3] gives real dimension $n(n+2)$ for trivial spherical monodromy and dimension n for dense circle monodromy. For dense PSU(2) monodromy, the family consists only of the tuple (4.10).

For a nonconstant meromorphic map $h : X \to \mathbb{CP}^1$, the pullback of the round metric represents the ramification divisor R_h. It therefore produces Toda metric tuples representing $(R_h, \dots, R_h)$ on every compact Riemann surface. These include examples with $b_i \in 4\pi\mathbb{N}$ for every i, but the construction does not prescribe an arbitrary divisor on a fixed X.

Toric solutions and character ensembles. Mu–Shi–Xu [4] call a Toda solution *toric* when its associated curve has monodromy in a maximal torus of PSU(N). They describe these solutions using *character n-ensembles* of meromorphic one-forms. In diagonal coordinates, the curve is

$$f = \left[1 : r_1 \exp\int \Omega_1 : \cdots : r_n \exp\int \Omega_n\right], \qquad r_j > 0. \tag{4.12}$$

The Ω_j have at most simple poles and purely imaginary periods, and the curve must be linearly full. Its osculating pullbacks (1.11) yield the metric tuple. The purely imaginary periods ensure unitary monodromy, so the metrics descend. Osculating ramification points are included in the singular set. The differentials are recovered from logarithmic derivatives of coordinate ratios. For each admissible ensemble, varying the positive constants gives an n-parameter family of metric tuples representing the same ordered divisors.

The component divisors depend on poles as well as on zeros of osculating Wronskians, so residues alone do not prescribe them. The monodromy group of a reduced solution need not be abelian, and the associated curve of a toric solution need not lie on a rational normal curve.

5 Open problems

The conformal structure and marked points are fixed unless they are explicitly varied.

1. **Existence for prescribed component divisors.** For a fixed marked compact surface (X, Σ), characterize the tuples $(D_1, \dots, D_n)$ admitting solutions of the Toda curvature system. The positivity condition (4.1) is not sufficient in general, and the noncriticality assumption $b_i \notin 4\pi\mathbb{N}$ in Theorem 4.2 is not necessary for existence. A general criterion must account for critical parameters, real weights, and compatibility of local exponents with global unitary monodromy. A criterion for arbitrary singularity data at three points on the sphere would extend the hypergeometric results of [10].
2. **Several plane sources and prescribed behavior at infinity.** Classify finite-energy solutions with several finite singularities and a fixed conical weight at infinity, equivalently solutions on the sphere with divisors (3.9). Determine the accessory parameters compatible with unitary monodromy. For solutions with polynomial energy growth, determine which rational or Laurent-polynomial ODE coefficients give projective unitary monodromy and which asymptotic or Stokes data are realized by Toda metrics. Extend the ODE descriptions of [8, 9] to existence criteria with prescribed asymptotic data.
3. **Classification and counting on compact surfaces.** Describe the dimension, connected components, and singular strata of solution spaces for fixed divisors, and determine when these spaces are finite or positive-dimensional. Extend the torus counting results to critical parameters and more general weights, and determine when the known upper bounds are attained as the modulus and marked points vary [13, 14, 15]. Use the correspondence with equivariant curves to obtain effective criteria and counts.

4. **Realizing and distinguishing geometric subclasses.** Determine which divisor tuples arise from character ensembles and describe the moduli of those ensembles [4]. For reduced solutions, determine how the existence and moduli of the underlying spherical metrics constrain the higher-rank family [5]. Describe the loci of toric solutions, reduced solutions, and solutions with irreducible monodromy, and their intersections and degenerations. Irreducibility means that the monodromy preserves no nonempty proper projective linear subspace. This property differs from linear fullness of the associated curve.
5. **Compactification, collisions, and critical limits.** Understand limits as marked points collide, the complex structure degenerates, or parameters approach the exceptional set for compactness. Construct a geometric compactification recording limiting unitary curves and bubbles, and relate analytic concentration data to changes in ramification divisors and degenerations of monodromy.

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